

COPPER paper source fallback render

```
\documentclass[10pt]{article}
\usepackage[margin=0.75in]{geometry}
\usepackage{url}
\usepackage{graphicx}
```

```
\title{COPPER: Committed Pointer-Provenance Prefetching}
\author{Artifact draft}
\date{}
```

```
\begin{document}
\maketitle
```

```
\begin{abstract}
```

COPPER explores whether committed pointer-provenance signals can improve the selectivity of hardware data-derived prefetching for pointer-intensive workloads. The mechanism records pointer-source evidence only after architectural commit, invalidates or blocks stale and mismatched sources, and uses that evidence to gate later prefetch issue. The artifact now supports a careful claim: COPPER is a measurable research mechanism with executable-model evidence, deterministic cycle-model evidence, deterministic core-integrated evidence, source-backed independent-simulator evidence, validated imported gem5 ARM-system rows when PASS rows exist, CI-proven RTL unit simulation, CI-proven paper/artifact reproduction, matched unit-level synthesis evidence, near-core-stub generic-resource evidence, PicoRV32 core-wrapper generic-resource evidence, matched near-core-stub and PicoRV32 tiny-SoC full-core mapped-FPGA PPA when PASS rows exist, scoped PicoRV32 core-wrapper OpenROAD post-route, ASIC-Liberty, or Vivado FPGA tool-estimated power when indexed PASS, and assumption-based proxy-energy rows. It does not claim state-of-the-art performance, production readiness, a complete gem5 workload matrix, production full-system integration, production-core mapped timing, measured silicon power efficiency, ASIC signoff, foundry signoff, or silicon signoff.

```
\end{abstract}
```

```
\section{Introduction}
```

Pointer-heavy programs expose memory latency that ordinary address-stream prefetchers can miss. Data-derived prefetching can help, but it can also act on data that merely resembles an address. COPPER changes the authority rule: a prefetch candidate needs committed pointer-source evidence before issue. This paper version is claim-disciplined; all measurements are routed through generated CSVs and all unsupported claims are kept out of the body.

```
\section{Motivation}
```

```
\begin{figure}[t]
```

```
\centering
```

```
\fbox{\begin{minipage}{\linewidth}\centering
```

Demand execution uses a loaded pointer after commit. COPPER records that source word as eligible. Later data-derived prefetch issue checks the same source identity, value evidence, domain, and permission path before acting.

```
\end{minipage}}
```

```
\caption{Pointer-chasing motivation.}
```

```
\label{fig:motivation}
```

```
\end{figure}
```

The motivating failure mode is not ordinary speculation alone. It is that a prefetcher may treat address-shaped data as authority. COPPER instead treats committed pointer use as authority.

```
\section{Background And Related Work}
```

Prior work covers next-line, stride, stream, indirect, pointer-chase, runahead, helper-thread, and dependence-guided prefetching, as well as metadata and capability mechanisms. COPPER's distinction is narrower: committed source-word proof gates data-derived prefetch issue. The repository matrix records what COPPER does not claim.

```
\section{COPPER Design}
```

```
\begin{figure}[t]
```

```
\centering
```

```
\fbox{\begin{minipage}{\linewidth}\centering
```

Commit path: observe pointer-source use. Provenance table: retain clean source evidence. Prefetch path: require source match, target permission, and valid translation context. Invalidation path: clear source evidence on writes and coherence events.

```
\end{minipage}}
```

```
\caption{COPPER architecture.}
```

```
\label{fig:architecture}
```

```
\end{figure}
```

```
\begin{figure}[t]
```

```

\centering
\fbbox{\begin{minipage}{\linewidth}\centering
Load source word, use as demand address, commit, record proof, later prefetch candidate, check proof, issue or
block.
\end{minipage}}
\caption{Committed provenance update timeline.}
\label{fig:timeline}
\end{figure}

```

COPPER has three core rules: proof creation occurs after committed architectural evidence; writes, coherence updates, failed permissions, and context mismatch block or destroy proof; recursive issue does not gain authority merely because a line arrived by prefetch.

```

\section{Implementation}
The artifact contains a Python model, a trace-driven evaluation harness, SystemVerilog RTL units, C/C++ workload
sources, reproduction scripts, and summary parsers. The open-source RTL smoke target is
\texttt{copper\_prefetch\_unit\_open}; larger local Vivado summaries are treated as existing evidence rather than
portable rerun proof.

```

```

\section{Methodology}
\begin{table}[t]
\centering
\caption{Benchmarks}
\begin{tabular}{lll}
\hline
Class & Examples & Evidence file \\
\hline
Pointer-heavy & linked list, tree, hash, graph, Patricia & workload\_build.csv; independent\_sim\_performance.csv \\
\hline
Controls & array, matrix, compute, random access & workload\_build.csv; independent\_sim\_performance.csv \\
Stress & short chains, long chains, mixed, noisy, branchy & workload\_build.csv; independent\_sim\_performance.csv \\
\hline
\end{tabular}
\end{table}

```

```

\begin{table}[t]
\centering
\caption{Configurations And Baselines}
\begin{tabular}{lll}
\hline
Config & Role & Evidence file \\
\hline
no\_prefetch & reference & cycle\_performance.csv \\
next\_line & spatial baseline & cycle\_performance.csv \\
stride & regular-stream baseline & cycle\_performance.csv \\
simple\_pointer\_chase & unsafe content-derived baseline & cycle\_performance.csv \\
copper & committed-provenance policy & cycle\_performance.csv \\
\hline
\end{tabular}
\end{table}

```

The normalized CSVs report per-workload rows rather than hiding negative results inside averages. Evidence levels are explicit: `\texttt{model}` rows come from the executable policy model, `\texttt{cycle\_model}` rows come from a deterministic memory-system model with hit/miss latency, memory latency, outstanding prefetches, queue drops, lateness, and demand/prefetch traffic accounting. `\texttt{core\_integrated}` rows add a deterministic core envelope with fetch/issue width, reorder-window pressure, load-queue pressure, branch penalties, and memory-system timing. `\texttt{independent\_sim}` rows execute the source-built C workload driver for checksums and then use a separate trace/event cache simulator that does not import the cycle-model or core-integrated harness. `\texttt{gem5\_full\_system}` rows are imported from validated ARM-system summaries only when the compared group has a no-prefetch baseline, a COPPER-family row, matching checksums, zero return codes, and positive tick counts. `\texttt{gem5\_statistical\_summary.csv}` reports imported-summary confidence intervals where repeated samples exist and marks single-sample rows explicitly. This paper does not claim a clone-local rerun of every raw simulator run. None of these rows claim complete CPU integration.

```

\section{Evaluation}
\begin{figure}[t]
\centering
\fbbox{\begin{minipage}{\linewidth}\centering

```

Speedup rows are generated in \texttt{research/results/performance.csv}, \texttt{research/results/cycle_performance.csv}, \texttt{research/results/core_integrated_performance.csv}, and \texttt{research/results/independent_sim_performance.csv}. The paper does not collapse them into a single broad win claim.

\end{minipage}}
\caption{Speedup by benchmark.}
\label{fig:speedup}
\end{figure}

\begin{figure}[t]
\centering
\fbbox{\begin{minipage}{\linewidth}\centering
Issued, useful, useless, accuracy, coverage, lateness, and queue-drop fields are generated in \texttt{research/results/prefetch_metrics.csv}, \texttt{research/results/cycle_prefetch_metrics.csv}, \texttt{research/results/core_integrated_prefetch_metrics.csv}, and \texttt{research/results/independent_sim_prefetch_metrics.csv}.
\end{minipage}}
\caption{Prefetch accuracy and coverage.}
\label{fig:prefetch}
\end{figure}

\begin{figure}[t]
\centering
\fbbox{\begin{minipage}{\linewidth}\centering
Demand-load, prefetch-load, total-request, and traffic-overhead fields are generated in \texttt{research/results/memory_traffic.csv}, \texttt{research/results/cycle_memory_traffic.csv}, \texttt{research/results/core_integrated_memory_traffic.csv}, and \texttt{research/results/independent_sim_memory_traffic.csv}.
\end{minipage}}
\caption{Traffic overhead.}
\label{fig:traffic}
\end{figure}

Current evidence supports selective, per-workload discussion. Several cycle-model, core-integrated, and independent-sim rows show COPPER behind the best baseline; those regressions remain in the CSVs and block a universal speedup claim.

\section{Ablation And Sensitivity}
\begin{figure}[t]
\centering
\fbbox{\begin{minipage}{\linewidth}\centering
Ablation and sensitivity rows are generated in \texttt{research/results/ablation.csv} and \texttt{research/results/sensitivity.csv}; the rows include evidence-level labels.
\end{minipage}}
\caption{Ablation and sensitivity.}
\label{fig:ablation}
\end{figure}

The cycle-model ablations isolate no provenance, speculative provenance, committed-only proof, confidence, queue filtering, and full COPPER. Queue size, confidence threshold, chain depth, distance, table size, and memory latency are varied in sensitivity rows. These are cycle-model counters and should not be described as gem5 counters.

\section{Hardware Cost}
\begin{table}[t]
\centering
\caption{Hardware Cost}
\begin{tabular}{lll}
\hline
Evidence & Scope & File \\
\hline
CI RTL simulation & open-source unit test & rtl_simulation.csv \\
Yosys & generic unit resource context & synthesis.csv \\
nextpnr & mapped unit resource context when available & synthesis.csv \\
Overhead & matched unit rows from same flow & synthesis_overhead.csv \\
Near-core stub & matched generic near-core-stub rows when Yosys runs & fullcore_synthesis_overhead.csv \\
PicoRV32 core-wrapper & matched generic accepted core-wrapper rows when Yosys runs & fullcore_synthesis_overhead.csv \\
PicoRV32 tiny-SoC full-core & matched full-core rows when Yosys or mapped flows run & fullcore_synthesis_overhead.csv; mapped_ppa.csv \\
\end{tabular}

Mapped PPA & matched near-core-stub, PicoRV32 core-wrapper, PicoRV32 tiny-SoC full-core, or unit rows only when place-and-route succeeds & mapped_ppa.csv; mapped_ppa_overhead.csv \\

Production-core integration & not claimed & top_tier_gate_status.csv \\

\\hline

\\end{tabular}

\\end{table}

The artifact supports unit-level hardware plausibility, matched near-core-stub resource-overhead rows, matched PicoRV32 accepted-core-wrapper rows, and matched PicoRV32 tiny-SoC full-core rows when the generated ledgers mark them PASS. The near-core stub is not a full CPU, the accepted wrapper is not full-core, and the PicoRV32 tiny-SoC full-core harness is not a production ARM/OoO integration. Generic Yosys rows do not provide mapped timing, Fmax, ASIC area, or measured power. Mapped timing may be discussed only when \\texttt{mapped_ppa.csv} contains matched PASS rows from nextpnr, Vivado, or OpenROAD.

\\section{Energy Proxy}

\\begin{table}[t]

\\centering

\\caption{Energy/Power Evidence}

\\begin{tabular}{ll}

\\hline

Evidence & Scope & File \\

\\hline

Memory traffic proxy & assumption-based, not measured & energy_proxy.csv \\

ASIC Liberty tool-power index & Nangate45 standard-cell estimate when indexed PASS & asic_power.csv;

power_report_index.csv \\

OpenROAD post-route power index & Nangate45 post-route tool estimate with indexed final DEF/SPEF/netlist when PASS & openroad_postroute_power.csv; power_report_index.csv \\

FPGA tool-power index & Vivado FPGA tool estimate when indexed PASS & power_report_index.csv; mapped_ppa.csv \\

Activity proxy & McPAT proxy when indexed PASS & power_report_index.csv; copper_mcpat_sensitivity_20260618.csv \\

Summary & proxy overhead statistics & energy_summary.csv \\

\\hline

\\end{tabular}

\\end{table}

Energy rows use explicit assumptions recorded in \\texttt{energy_proxy.csv}. When \\texttt{power_report_index.csv} marks \\texttt{openroad_postroute_tool_estimate} PASS, the row is a Nangate45 OpenROAD-flow-scripts post-route estimate with indexed final DEF/SPEF/netlist artifacts, not silicon measurement or foundry signoff. When \\texttt{asic_liberty_tool_estimate} is PASS, the row is a Nangate45 standard-cell Liberty estimate from OpenSTA/OpenROAD, not silicon measurement or post-route signoff with extracted parasitics. When \\texttt{fpga_tool_estimate} is PASS, the row is Vivado \\texttt{report_power} for the mapped FPGA target, not silicon or ASIC signoff. When \\texttt{proxy_activity} is PASS, the activity proxy is the fixed-architecture McPAT sensitivity run driven by measured gem5 ROI counters. This supports scoped tool-power and proxy/model energy discussion, not foundry-signoff or silicon power-efficiency claims.

\\section{Limitations}

\\begin{table}[t]

\\centering

\\caption{Limitations}

\\begin{tabular}{ll}

\\hline

Limitation & Consequence \\

\\hline

Gem5 rows are validated imported ARM-system summaries & Gem5 validation is scoped to imported PASS rows \\

Independent simulator is trace/event level & It is not a replacement for a broad gem5 campaign \\

PicoRV32 tiny-SoC is the scoped full-core target & Production ARM/OoO claims remain blocked \\

Power is tool-estimate/model based & Signoff and silicon power-efficiency claims remain blocked \\

External gem5 and Vivado setup & Large external-tool reruns are not clone-local \\

\\hline

\\end{tabular}

\\end{table}

\\section{Artifact Evidence Gates}

\\begin{table}[t]

\\centering

\\caption{Artifact And Evidence Gates}

\\begin{tabular}{lll}

\\hline

Gate class & Status source & Interpretation \\

\hline CI RTL evidence & rtl_simulation.csv & GitHub Actions unit simulation \\ Cycle-model evidence & cycle_performance.csv & deterministic memory-system timing model \\ Core-integrated evidence & core_integrated_performance.csv & deterministic core-envelope model \\ Independent simulator & independent_sim_performance.csv & source-backed trace/event simulator \\ Gem5 validated summaries & gem5_validation.csv; gem5_performance.csv; gem5_statistical_summary.csv & imported ARM-system rows and imported-summary statistics when PASS \\ Hardware cost & synthesis_overhead.csv; fullcore_synthesis_overhead.csv; mapped_ppa.csv & matched unit, near- core-stub, PicoRV32 core-wrapper, or PicoRV32 tiny-SoC full-core resources, plus mapped timing only when PASS \\ Energy proxy & energy_proxy.csv; openroad_postroute_power.csv; asic_power.csv; power_report_index.csv & assumption-based proxy plus OpenROAD post-route, ASIC Liberty, or FPGA tool estimates when indexed PASS \\ Package & artifact_manifest.csv & included and excluded files \\ \hline \end{tabular}
--

\section{Conclusion}
COPPER is best framed as a committed-provenance authority mechanism for data-derived prefetch issue. The artifact now has a CI-proven open-source path, deterministic cycle-model and core-integrated evidence, source-backed independent-simulator rows, validated imported gem5 ARM-system rows with imported-summary statistics, matched unit-level, near-core-stub, PicoRV32 accepted-core-wrapper, and PicoRV32 tiny-SoC full-core resource/PPA paths, FPGA tool-estimated power where indexed PASS, proxy energy rows, explicit blocked OpenROAD/ASIC-Liberty entries where those reports are absent, a claim ledger, audits, and a package manifest. The honest next step for a stronger architecture submission is a final raw-rerunnable gem5 or comparable external core validation matrix plus signoff-calibrated or silicon-measured power evidence without changing the scoped claims in this artifact.

\bibliographystyle{plain}
\bibliography{references}
\end{document}